

Module 3: Data Organization and Manipulation using Collections

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In module 3,

- Data Handling Strategies Using Lists, Tuples, Dictionary and Set
- Data Comprehension
- Data Modification
- Grouping and Categorization
- Iterators
- Data Selection - Ordered, Unordered and Unique Data
- Searching Techniques: Linear Search, Binary Search
- Sorting Techniques: Bubble Sort, Selection Sort- Insertion Sort- Quick Sort- Merge Sort.

What is a Set?

- A set is an unordered collection of unique elements.
- Defined using curly braces or `set()` constructor.
- Duplicates are automatically removed.
- Unordered → No index, no slicing.
- Useful for membership testing, mathematical set operations (union, intersection, difference).

Characteristics of Set

- Unordered → Elements don't have a fixed position.
- Unique → No duplicate elements allowed.
- Mutable → Can add/remove elements.
- Heterogeneous → Can store mixed data types, but only immutable ones (int, str, tuple).
- No indexing/slicing like lists/tuples.

Set creation

Example (Syntax)

```
s1 = set() # Empty set
```

```
s2 = {1, 2, 3, 4}
```

```
# From list (duplicates auto-removed)
```

```
s3 = set([1, 2, 2, 3, 4, 4])
```

```
print(s3) # {1, 2, 3, 4}
```

```
# Mixed datatypes
```

```
s4 = {"apple", 10, 3.14, (1,2)}
```

Set Operations (Mathematical)

Example (Syntax)

$A = \{1, 2, 3, 4\}$

$B = \{3, 4, 5, 6\}$

`print(A | B)`

`print(A & B)`

`print(A - B)`

`print(A ^ B)`

Example (Syntax)

```
s = {1, 2, 3}  
s.add(4)  
print(s)
```

```
s = {1, 2}  
s.update([3, 4], {5, 6})  
print(s)
```

```
s = {1, 2, 3}  
s.remove(2)  
print(s)
```

Set Methods

Example (Syntax)

```
s = {1, 2, 3}
s.discard(5)
print(s)
```

```
s = {10, 20, 30}
x = s.pop()
print("Removed:", x)
print("Remaining:", s)
```

```
s = {1, 2, 3}
s.clear()
print(s)
```

Example (Syntax)

```
s = {1, 2, 3}  
c = s.copy()  
print(c)
```

Mathematical Set Operations

Example (Syntax)

```
A = {1, 2, 3}
```

```
B = {3, 4, 5}
```

```
print(A.union(B))
```

```
print(A | B)
```

```
A = {1, 2, 3}
```

```
B = {2, 3, 4}
```

```
print(A.intersection(B))
```

```
print(A & B)
```

Mathematical Set Operations

Example (Syntax)

```
A = {1, 2, 3}
B = {2, 3, 4}
print(A.difference(B))
print(B - A)
```

```
A = {1, 2, 3}
B = {3, 4}
print(A.symmetric_difference(B))
print(A ^ B)
```

Subset, Superset & Disjoint Checks

Example (Syntax)

```
A = {1, 2}
B = {1, 2, 3, 4}
print(A.issubset(B))
```

```
A = {1, 2, 3, 4}
B = {2, 3}
print(A.issuperset(B))
```

```
A = {1, 2}
B = {3, 4}
print(A.isdisjoint(B))
```

Set Comprehension

Example (Syntax)

```
{expression for item in iterable if condition}
```

Set Comprehension

Example (Syntax)

```
squares = {x**2 for x in range(6)}  
print(squares)
```

Set Comprehension

Example (Syntax)

```
evens = {x for x in range(10) if x % 2 == 0}  
print(evens)
```

Example (Syntax)

```
chars = {ch for ch in "programming"}  
print(chars)
```

Set Comprehension

Example (Syntax)

```
multiples_of_3 = {x for x in range(20) if x % 3 == 0}  
print(multiples_of_3)
```

Set Comprehension

Example (Syntax)

```
matrix = [[1, 2], [3, 4], [5, 6]]  
flattened = {num for row in matrix for num in row}  
print(flattened)
```

Example (Syntax)

```
words = ["apple", "banana", "cherry"]  
lengths = {len(word) for word in words}  
print(lengths)
```

Set Comprehension

Example (Syntax)

```
A = {1, 2}
B = {3, 4}
pairs = {(a, b) for a in A for b in B}
print(pairs)
```

Questions

- Create a set with numbers 1–5 and display it.
- Add an element 10 to a set.
- Remove an element from a set using both `remove()` and `discard()`.
- Create a set from a string "hello" and print it.
- Convert a list with duplicate values `[1,2,2,3,3,4]` into a set and explain the result.

Questions

- Given $A=1,2,3,4,5$ and $B=4,5,6,7,8$, find elements present in exactly one of the sets.
- Remove all duplicates from a list of 100 random numbers using a set.
- Given a sentence, extract all unique words using a set.
- Find all common characters between two strings using sets.
- Write a program to check if two strings are anagrams using sets.

Questions

- Generate a set of all numbers between 1–50 that are divisible by both 3 and 5.
- Extract all unique vowels from the sentence "Python programming is fun".
- Flatten a nested list `[[1,2,2],[3,4,4],[5,6]]` into a unique set of numbers.
- Create a set of tuples (x, x^2) for x in range 1–10.
- From a list of words `["apple", "banana", "pear", "peach", "plum"]`, create a set of first letters.
- Create a set of all prime numbers under 50 using comprehension + function.
- Generate all unique pairs (a,b) where $a \in 1,2,3$ and $b \in x,y$, but exclude pairs where $a==2$.